

Joongwon Chae

Master's Student Researcher · Training-Free Anomaly Detection · Memory-Augmented Inference · Medical AI

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RESEARCH INTERESTS

I develop training-free and gradient-free inference methods that adapt frozen foundation-model representations through memory, retrieval, prompting, routing, and calibration. My work centers on anomaly detection and medical image analysis under limited labels, retraining budgets, and human interaction.

EDUCATION

Tsinghua University - Shenzhen International Graduate School (SIGS) 2024-2027 (expected)
M.S. in Electronic Information (Biomedical Engineering), Institute of Biopharmaceutical and Health Engineering · Advisor: Prof. Peiwu Qin · Shenzhen, China

Shanghai Jiao Tong University (SJTU) - Department of Automation 2018-2024
B.S. in Automation · Shanghai, China

RESEARCH EXPERIENCE

Master's Student Researcher - Tsinghua University (SIGS) 2024-present
Independently leading method development under the supervision of Prof. Peiwu Qin.

- **Training-free anomaly detection (ProCon, GCR, StructCore)**. Led three first-author works end-to-end: research questions, method design and implementation, experiments, ablations, and manuscript writing. ProCon reports 99.8%, 99.2%, and 93.2% image AUROC on MVTec AD, VisA, and Real-IAD with a frozen backbone.
- **Human-prompt-free medical segmentation (Memory-SAM)**. Designed retrieval-to-prompt inference that converts frozen DINOv3 correspondence into automatic SAM2 prompts. The public v3 preprint reports 0.9863 mIoU on a mixed expert-annotated test split without manual clicks or fine-tuning.
- Owned the full research cycle from problem formulation through open-source release.

Clinical ML study - gallbladder-cancer T-staging 2024-2025
First-authored the Frontiers in Oncology study; designed classical ML baselines, SMOTE analysis, AUROC evaluation, and 1,000-sample bootstrap confidence intervals across 232 patients.

Multimodal medical-AI systems (TCM) - co-first author 2024-present
Co-led ViTCM-LLM and MMIR-TCM; co-developed MedTCM, designed the TDEU evaluation metric, and contributed retrieval-augmented reasoning and safety constraints.

PEER-REVIEWED PUBLICATIONS

1. **Joongwon Chae**, Zhenyu Wang, Duanpo Wu, Lian Zhang, Alexander Tuzikov, Magrupov Talat Madiyevich, Min Xu, Dongmei Yu, Peiwu Qin. "Pre-operative T-stage discrimination in gallbladder cancer using machine learning and DeepSeek-R1." *Frontiers in Oncology*, 15, Article 1613462, 2025. [link](#)
2. Lihui Luo*, **Joongwon Chae***, Yang Liu, Igor Pantic, Vladan Devedzic, Zhumei Sun, Zelin Zeng, Sanatbek Matlatipov, Xiaoming Yin, Peiwu Qin. "ViTCM-LLM: A Multimodal RAG Framework for Advanced TCM Clinical Decision Support." *2025 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, pp. 3894-3899. (* Equal contribution) [link](#) / [code](#)

PREPRINTS & MANUSCRIPTS

3. **Joongwon Chae**, Lihui Luo, Yang Liu, Dongmei Yu, Peiwu Qin, Runming Wang, Ilmoon Chae. "ProCon: Projection-Consistency Memory for Training-Free Anomaly Detection." *arXiv:2607.04894 [cs.CV]*, 2026. [link](#) / [code](#)
4. **Joongwon Chae**, Lihui Luo, Yang Liu, Runming Wang, Dongmei Yu, Zeming Liang, Xi Yuan, Dayan Zhang, Zhenglin Chen, Peiwu Qin, Ilmoon Chae. "GCR: Geometry-Consistent Routing for Task-Agnostic Continual Anomaly Detection." *arXiv:2601.01856 [cs.CV]*, 2026. [link](#) / [code](#)
5. **Joongwon Chae**, Lihui Luo, Yang Liu, Runming Wang, Dongmei Yu, Zeming Liang, Xi Yuan, Dayan Zhang, Zhenglin Chen, Peiwu Qin, Ilmoon Chae. "StructCore: Structure-Aware Image-Level Scoring for Training-Free Unsupervised Anomaly Detection." *arXiv:2602.17048 [cs.CV]*, 2026. [link](#) / [code](#)
6. **Joongwon Chae**, Lihui Luo, Xi Yuan, Dongmei Yu, Zhenglin Chen, Lian Zhang, Peiwu Qin. "Memory-SAM: Human-Prompt-Free Tongue Segmentation via Retrieval-to-Prompt." *arXiv:2510.15849 [cs.CV]*, 2025. [link](#) / [code](#)
7. Lihui Luo, **Joongwon Chae**, Ziyang Chen, Yang Liu, Siyi Cheng, Weihang Gao, Zelin Zeng, Xiaoming Yin, Samaneh Beheshti Kashi, Dongmei Yu, Lian Zhang, Jing Sui, Zeming Liang, Jiansong Ji, Peter E. Lobie, Peiwu Qin. "MMIR-TCM: Memory-Integrated Multimodal Inference and Retrieval for TCM Clinical Decision Support." *arXiv:2607.01814 [cs.AI]*, 2026. [link](#)
8. **Joongwon Chae**, Runming Wang, Chen Xiong, Gong Yunhan, Lian Zhang, Ji Jiansong, Dongmei Yu, Peiwu Qin. "Auditable Context-Aware HFMD Forecasting with Structured LLM Agents." *arXiv:2511.23276 [cs.LG, cs.MA]*, 2025. [link](#)
9. Lihui Luo*, **Joongwon Chae***. "ViGen: Video-based Generation with GRPO for Dynamic Image Editing." *Manuscript under review; co-first authors shown; full author list not yet public.*

HONORS & SCHOLARSHIPS

- **Excellent Student Scholarship, First Class** (International Students), Tsinghua University, 2024
- **Yingcai First-Class Scholarship**, Institute of Biopharmaceutical and Health Engineering, Tsinghua University, 2025
- **Tsinghua University International Graduate Tuition Scholarship**, 2025-2026
- **Shenzhen Universiade International Scholarship Foundation Scholarship**, 2026

INDUSTRY RESEARCH EXPERIENCE

Ratel Soft - Research Intern

Jul 2025-present

Developing vision-based anomaly-detection pipelines for collaborative-robot assembly-line inspection; benchmarking frozen foundation-model features and building PyTorch data and evaluation tooling.

TECHNICAL SKILLS

- **Programming:** Python, PyTorch, scikit-learn, OpenCV
- **Foundation models:** DINOv2/v3, SAM/SAM2, CLIP; Hugging Face Transformers, PEFT/LoRA, LLaMA-Factory
- **Systems:** Docker, DeepSpeed, multi-GPU training, RAG pipelines, industrial machine-vision cameras

LANGUAGES

Korean (native) · English (professional working proficiency) · Chinese (HSK 6) · Japanese (JLPT N2)